U.S. AIRLINE PERFORMANCE & DELAY ANALYSIS

Introduction

The aviation industry generates massive volumes of operational data every day, including flight schedules, delays, cancellations, airline performance, and airport operations. Analyzing this data helps airlines and airports improve operational efficiency, reduce delays, enhance customer satisfaction, and optimize resource allocation.

This project focuses on analyzing U.S. domestic flight data using MySQL Workbench, SQL, and Power BI to generate meaningful business insights. The dataset contains information related to flights, airlines, airports, delays, cancellations, and diversions. Through data cleaning, SQL analysis, and dashboard visualization, the project evaluates airline performance, airport performance, delay causes, and seasonal flight trends.

The insights generated from this analysis support data-driven decision-making for improving airline operations, reducing delays, and enhancing passenger experience.

Problem Statement

Airlines and airports often face challenges in understanding:

- Which airlines perform best in terms of delays and cancellations
- Which airports experience the highest traffic and delays
- Major causes of flight delays
- Monthly and seasonal flight trends
- Cancellation and diversion patterns
- On-Time Performance (OTP) of airlines
- Operational inefficiencies affecting customer satisfaction

Objectives

- Analyze flight performance across airlines
- Identify delay and cancellation patterns
- Evaluate airport operational performance
- Study monthly and weekly flight trends

• Calculate important aviation KPIs

• Develop interactive Power BI dashboards

• Generate business recommendations

Dataset Description

The final analytical dataset “flight_analysis” was created by integrating multiple aviation datasets.

The dataset contains flight transactions along with airline and airport information.

Datasets Used

Flights Dataset

Contains detailed flight records including:

- Flight Number
- Airline Code
- Origin Airport
- Destination Airport
- Departure Delay
- Arrival Delay
- Cancellation Information
- Diversion Information

Airlines Dataset

Contains:

- Airline Code
- Airline Name

Airports Dataset

Contains:

- Airport Code
- Airport Name
- City
- State

Important Columns

Column Name	Description
year	Flight Year
month	Flight Month
day	Flight Day
day_of_week	Day of Week
airline	Airline Code
flight_number	Flight Number
origin_airport	Source Airport
destination_airport	Destination Airport
departure_delay	Departure Delay
arrival_delay	Arrival Delay
cancelled	Cancellation Status
diverted	Diversion Status
cancellation_reason	Cancellation Reason
airline_delay	Airline Delay Minutes
weather_delay	Weather Delay Minutes
air_system_delay	Air System Delay Minutes
security_delay	Security Delay Minutes
late_aircraft_delay	Late Aircraft Delay Minutes
airline_name	Airline Name
origin_city	Origin City
destination_city	Destination City

Tools Used

Tool / Technology	Purpose
MySQL Workbench	Database management and SQL execution
SQL	Data cleaning, transformation, and analysis
MySQL	Data storage and management
Power BI	Dashboard development and visualization
CSV Files	Source datasets

Data Cleaning and Preparation

The following preprocessing steps were performed:

• Imported flight, airline, and airport datasets into MySQL

• Verified column names and data types

• Handled missing values

• Created Flight Date column

• Created Cancellation Reason Description column

• Integrated airline and airport information with flight records

• Created a unified analytical dataset named flight_analysis

Feature Engineering

Flight Date

A new Flight Date column was created by combining Year, Month, and Day fields.

Cancellation Reason Description

Cancellation codes were converted into meaningful categories:

Code	Description
A	Airline / Carrier
B	Weather
C	National Air System
D	Security

SQL Analysis Summary

SQL queries were used to analyze airline operations and generate business insights.

Key Analyses Performed

Total Flights Analysis

Calculated total flight volume.

Cancellation Analysis

Calculated:

- Total Cancelled Flights
- Cancellation Rate

Diversion Analysis

Calculated:

- Total Diverted Flights
- Diversion Rate

Delay Analysis

Calculated:

- Average Arrival Delay
- Average Departure Delay

Airline Performance Analysis

Analyzed:

- Total Flights by Airline
- Average Delay by Airline
- OTP Rate by Airline
- Cancellation Rate by Airline

Airport Performance Analysis

Analyzed:

- Top 10 Busiest Airports
- Airport Delay Analysis
- Airport Cancellation Analysis

Delay Cause Analysis

Analyzed:

- Airline Delay
- Weather Delay

- Air System Delay
- Security Delay
- Late Aircraft Delay

Power BI Dashboard Overview

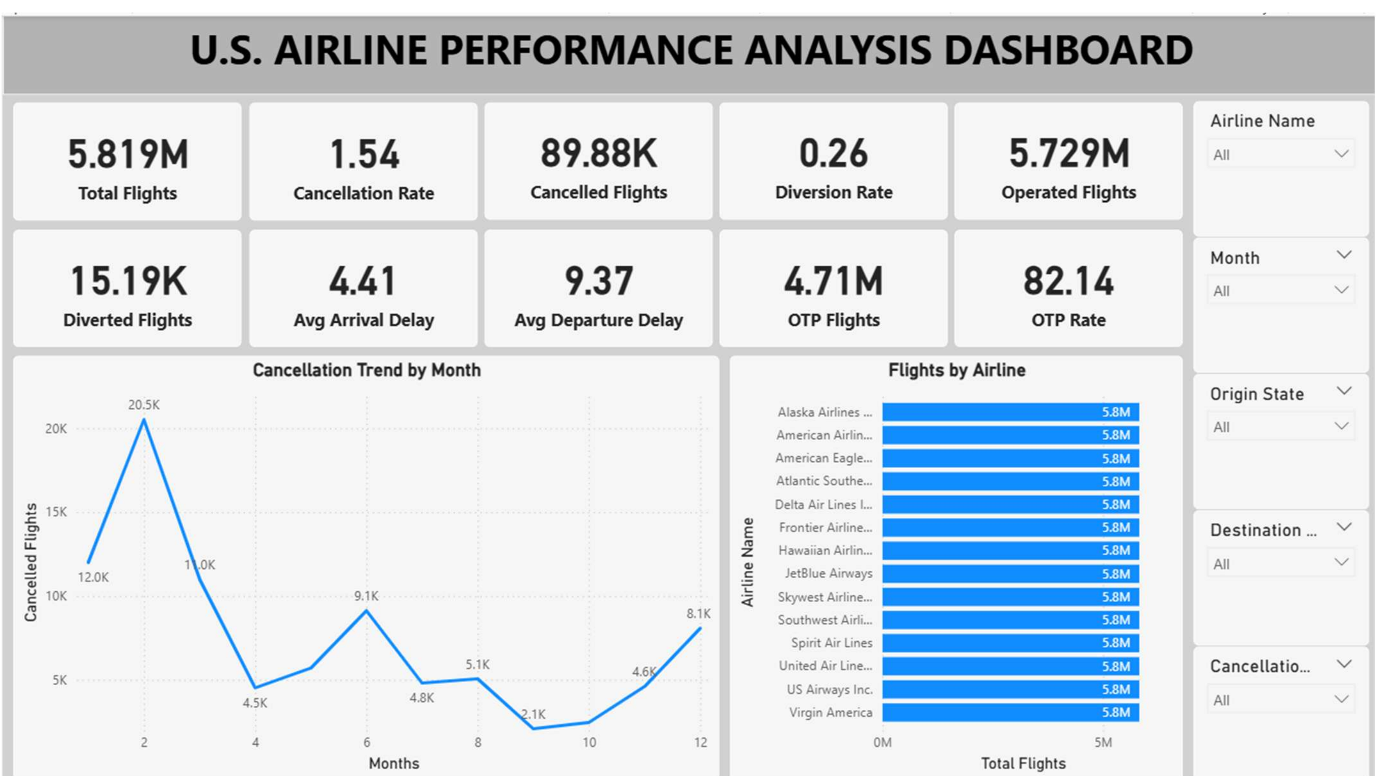
Page 1 – Executive Overview

KPIs:

- Total Flights
- Operated Flights
- Cancelled Flights
- Cancellation Rate
- Diversion Rate
- Average Arrival Delay
- Average Departure Delay
- OTP Flights
- OTP Rate

Visualizations:

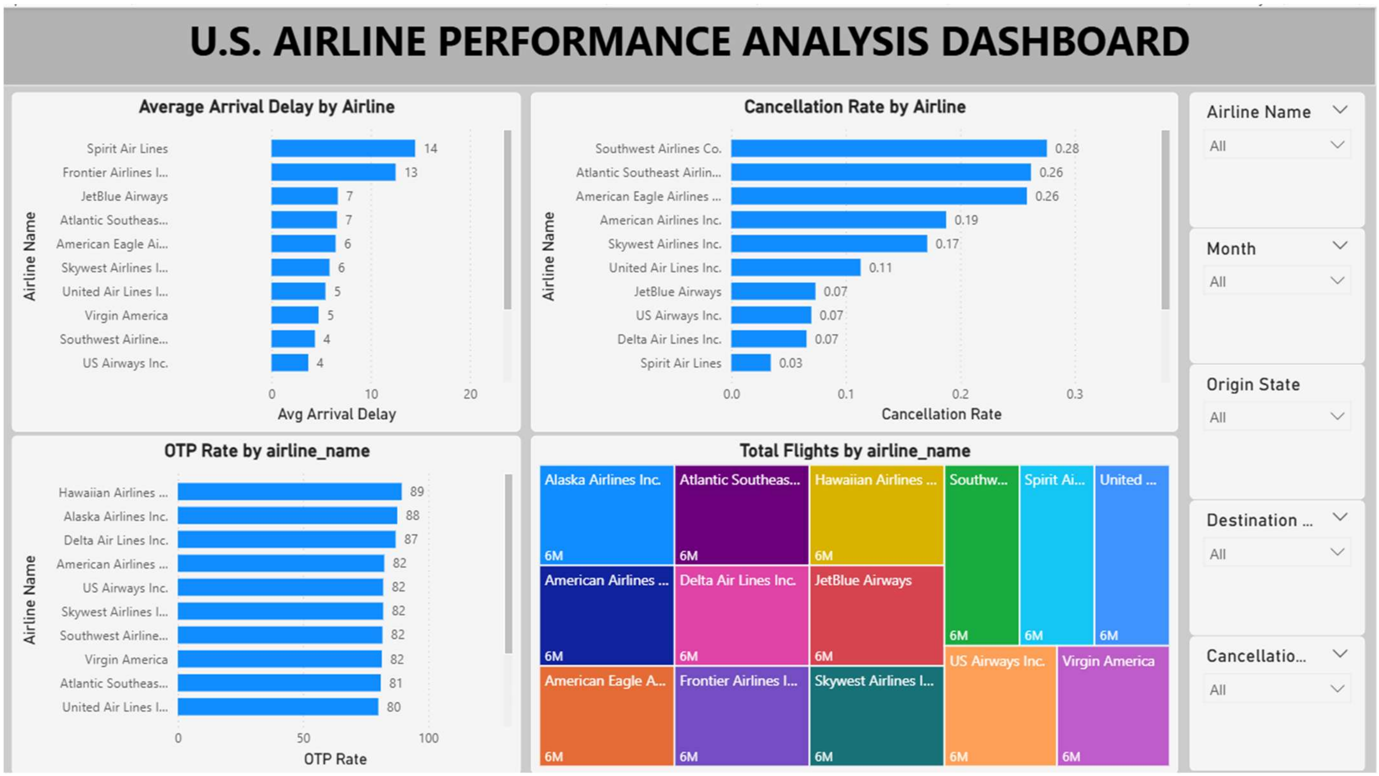
- Cancellation Trend by Month
- Flights by Airline



Page 2 – Airline Performance Analysis

Visualizations:

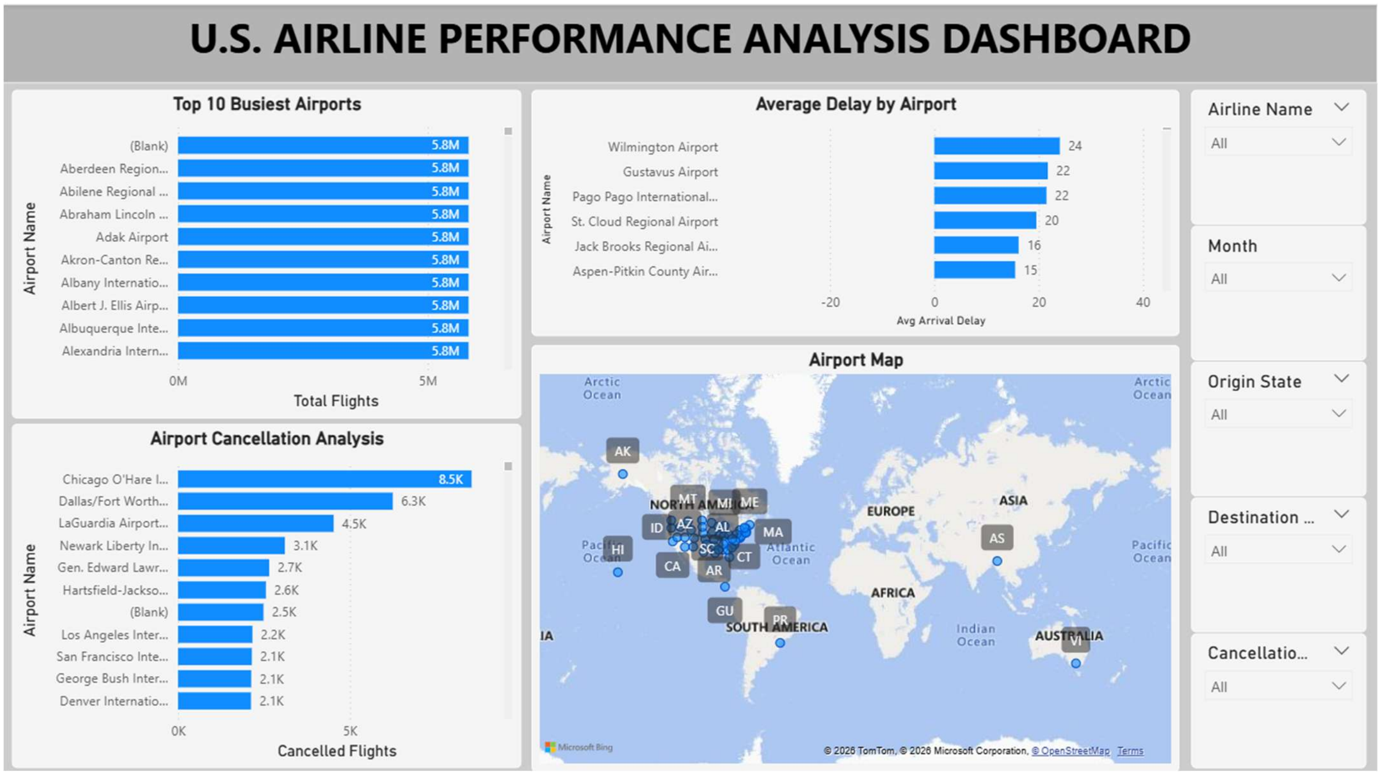
- Average Arrival Delay by Airline
- Cancellation Rate by Airline
- OTP Rate by Airline
- Total Flights by Airline



Page 3 – Airport Performance Analysis

Visualizations:

- Top 10 Busiest Airports
- Average Delay by Airport
- Airport Cancellation Analysis
- Airport Location Map



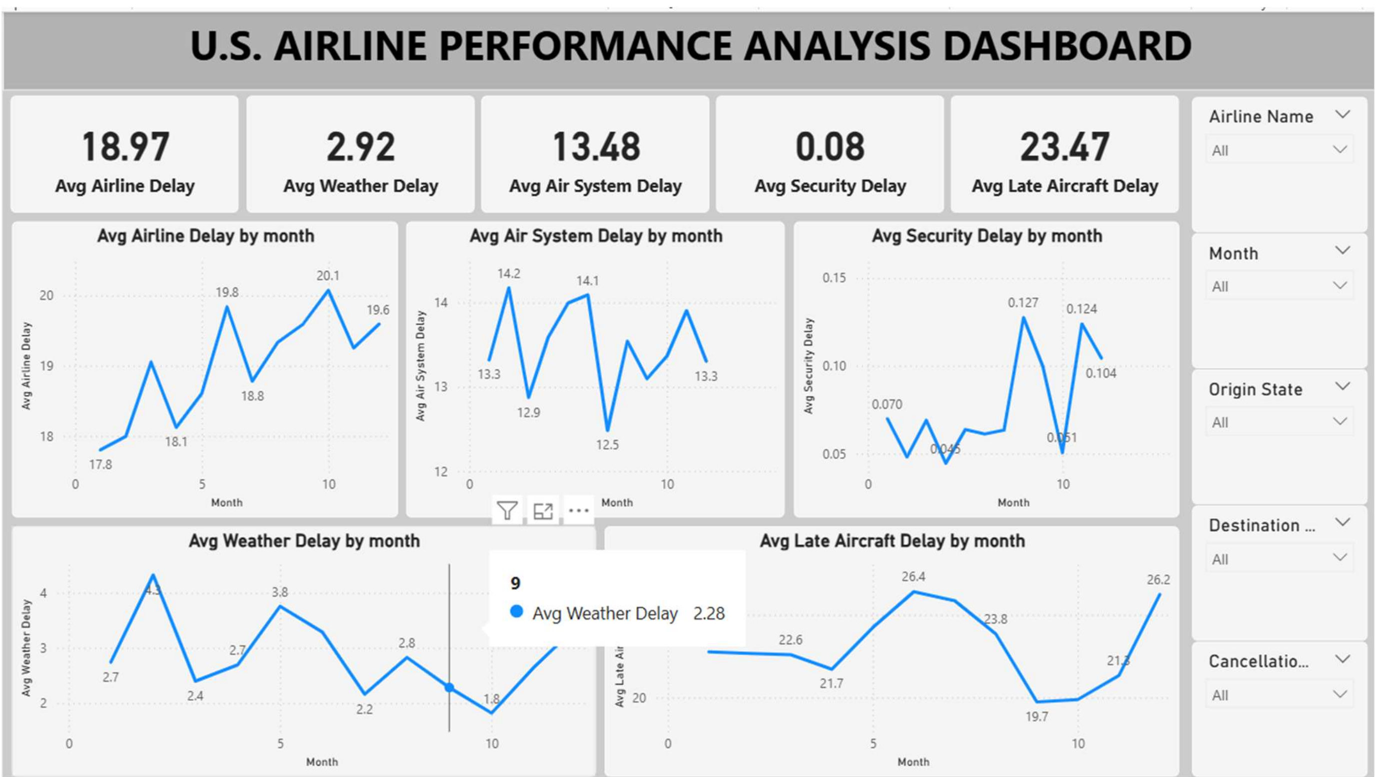
Page 4 – Delay Analysis

KPIs:

- Average Airline Delay
- Average Weather Delay
- Average Air System Delay
- Average Security Delay
- Average Late Aircraft Delay

Visualizations:

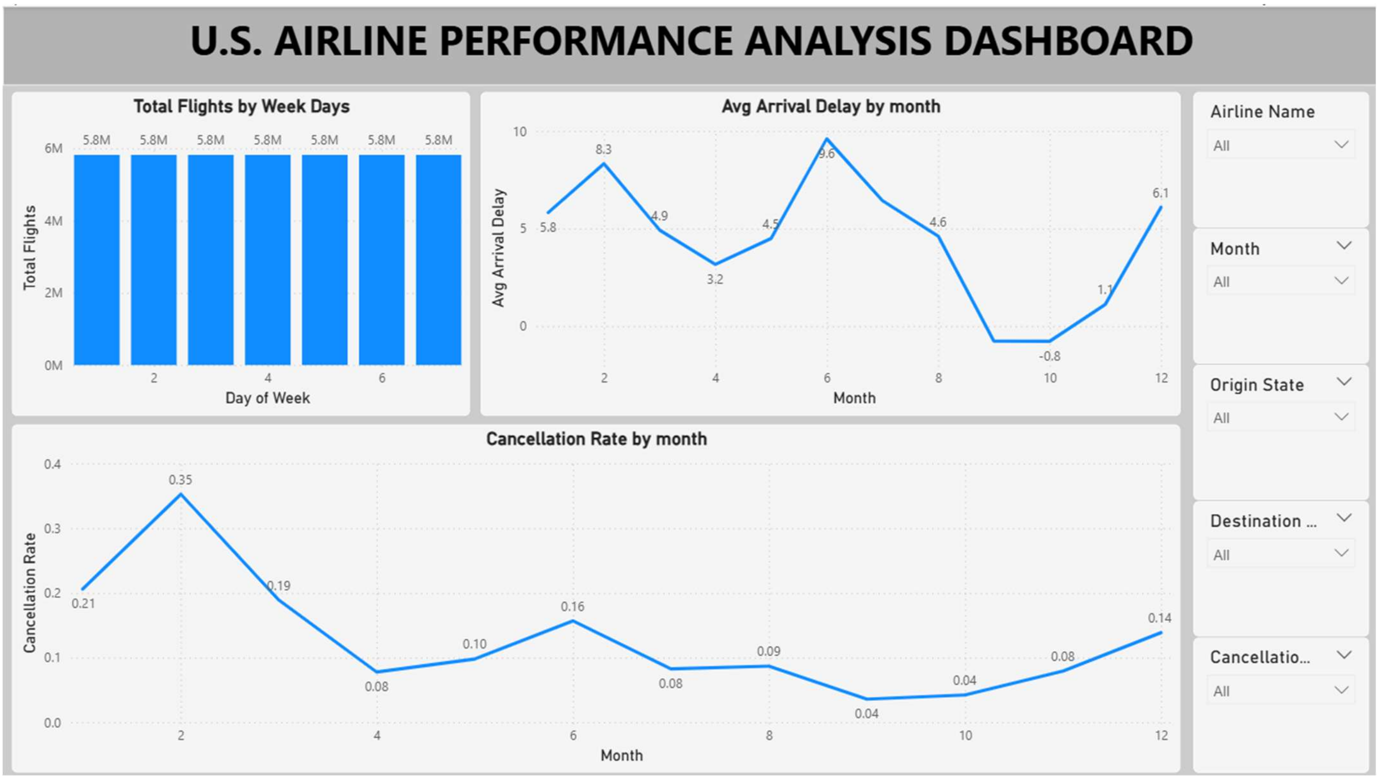
- Monthly Airline Delay Trend
- Monthly Weather Delay Trend
- Monthly Air System Delay Trend
- Monthly Security Delay Trend
- Monthly Late Aircraft Delay Trend



Page 5 – Time-Based Analysis

Visualizations:

- Flights by Day of Week
- Average Arrival Delay by Month
- Cancellation Rate by Month



Key Insights

- More than 5.8 million flight records were analyzed.
- The overall cancellation rate was approximately 1.54%.
- The diversion rate remained below 1%, indicating relatively stable operations.
- On-Time Performance exceeded 80%, demonstrating strong airline efficiency.
- Late Aircraft Delay contributed the highest average delay among all delay categories.
- Flight cancellations showed significant variation across months.
- Certain airlines consistently achieved higher OTP rates and lower delays.
- Some airports experienced significantly higher cancellation volumes.
- Weather-related delays increased during specific periods of the year.
- Major airports handled a large share of total flight traffic.
- Monthly flight trends revealed seasonal travel patterns.
- Interactive Power BI dashboards enabled efficient monitoring of airline operations.

Recommendations

• Improve aircraft turnaround processes to reduce late aircraft delays.

• Increase operational preparedness during adverse weather conditions.

• Monitor airline performance using OTP and delay KPIs.

• Optimize airport scheduling and resource allocation.

• Implement predictive maintenance programs to reduce operational disruptions.

• Develop contingency plans for high-cancellation periods.

• Use delay trend analysis to improve flight scheduling.

• Continuously monitor operational KPIs using Power BI dashboards.

Conclusion

This project successfully analyzed over 5.8 million U.S. domestic flight records using MySQL, SQL, and Power BI. The data was cleaned, transformed, integrated, and analyzed to generate valuable insights regarding airline performance, airport efficiency, delays, cancellations, and operational trends.

The Power BI dashboards provided a comprehensive and interactive view of key aviation metrics, enabling better understanding of operational performance and supporting data-driven decision-making. The findings and recommendations from this analysis can help airlines and airports improve efficiency, reduce delays, and enhance overall passenger satisfaction.